

Subject Description Form

Subject Code	ITC3223K
Subject Title	Circular and Warp Knitting for Fashion
Credit Value	3
Level	3
Pre-requisite(s) / <Co-requisite> / (Exclusion)	ITC2202T Foundations of Textiles I
Objectives	The subject enhances the knowledge of other types of knitting techniques, including circular and warp knitting techniques and machineries, to produce knitted items in three dimensional shape such as shocks, gloves and seamless garment.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - describe and elaborate the basic principles of fabric formation by circular and warp knitting techniques; - identify and compare the different knitted fabric structures and properties; - describe and elaborate the seamless knitted products by circular and warp knitting techniques; - communicate effectively with industrial practitioners; - develop a foundation for further study and engage in life-long learning; - identify and apply the safety and ethical practices in the industry.
Subject Synopsis/ Indicative Syllabus	(I) Non Jacquard Piece Goods by Circular Knitting Review of the basic principles of fabric formation on circular machines. Review of the yarns for the non-jacquard fabrics and the relationship between yarn type and fabric application. Basic requirements for the production of circular knitting: single jersey, double jersey, fleece, piques, and color striping. Fabric properties and production costing. (II) Jacquard Fabrics by Circular Knitting Machines and yarns for the production of single and double jacquard fabrics with the aid of computer pattern preparation systems, including multi-color pattern fabrics with various backings and the effect on fabric properties when using fiber blend, yarn blend and feeder blend techniques. (III) Warp Knit Fabrics Warp knit fabric classification and applications. Yarn requirement and machine types. Basic calculations for patterning, yarn weight and production costing. Study of lace fabrics for apparel and upholstery. Warp knit fabrics for industrial purposes.

**Reading List and
References**

Books

Brackenbury, T. (1992), *Knitted Clothing Technology*. Wiley-Blackwell, NJ.

Fukuhara: Precision Fukuhara Works, Ltd. (2006), *創立 50 周年記念冊：大圓机針織技術. Circular knitting technology*, Kabushiki Kaisha Seisakujo, Kobe-shi. 株式会社福原精機製作所, 神戸市.

Paling, D. F. (1965), *Warp knitting technology*. 2nd Edition, Columbine, Buxton England.

Raz, S. (1987), *Warp knitting production*. Melliand Textilberichte, Heidelberg.

Spencer, D. (2001), *Knitting technology a comprehensive handbook and practical guide*. 3rd edition, Woodhead publishing Ltd.